1.7 When Factorisation Fails

WORKSHEET

The question

STEP I 2016, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s1-q7

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps. There is space after each hint for your working.

Ask yourself List the first few members of S and T. Which members of S are *S-prime*, according to the definition in part (iv)?

STAGE 1 Union, intersection and products

Covers: Parts (i) and (ii)

Ask yourself What remainders on division by 4 do the members of S and T leave? Which positive integers can leave those remainders?

Ask yourself Multiply $(4a + 1)(4b + 1)$ and write the result as $4 \times$ (something) $+ 1$.

Pause and try For T, try a small example such as 3×3. It is also worth working out $(4a+3)(4b+3)$ and $(4a+1)(4b+3)$, since later parts use them.

STAGE 2 A prime factor in T

Covers: Part (iii)

Ask yourself The prime factors of a number in T are odd, so each lies in S or in T.

Pause and try What would happen if every prime factor lay in S? Use part (ii) repeatedly.

STAGE 3 An odd number of factors

Covers: Part (iv)(a)

Ask yourself Every member of T is at least 3. Use strong induction to show each $n \in T$ is T-prime or a product of T-primes.

Pause and try Why must the number of T-primes in the product be odd? Think about what a product of an even number of members of T leaves on division by 4.

STAGE 4 Two factorisations

Covers: Part (iv)(b)

Ask yourself Is 15 a T-prime? Is it prime? The same thing can happen in S: which members of S are products of two ordinary primes from T?

Pause and try Take two primes from T, say 3 and 7. With two copies of each, how many ways are there to pair them up into members of S?

What's really going on

What makes the integers special

Why *does* factorisation work in the ordinary integers? The key fact is *Euclid's lemma*: if a prime p divides a product ab, then p divides a or p divides b. It is proved using the Euclidean algorithm, and it is exactly what fails in S. There, the S-prime 9 divides $21 \times 21 = 441$ (because $441 = 9 \times 49$), but 9 doesn't divide 21 in S. In your first year at university you will prove Euclid's lemma carefully and see that unique factorisation follows from it.

The failed proof of Fermat's Last Theorem

On 1 March 1847 Gabriel Lamé announced to the Paris Academy that he had proved Fermat's Last Theorem. His idea was to factorise $x^n + y^n$ using complex roots of unity,

$$x^n + y^n = (x+y)(x+\zeta y)(x+\zeta^2 y)\cdots(x+\zeta^{n-1}y), \qquad \zeta = e^{2\pi \mathrm{i}/n}$$

(for odd n), and then argue about the prime factors of each bracket. Joseph Liouville immediately objected: why should numbers built from ζ factorise uniquely? Weeks later it emerged that Ernst Kummer had already shown that they often don't. The first failure is at $n = 23$.

Kummer rescued what he could by inventing "ideal numbers", extra objects that restore unique factorisation, and proved Fermat's Last Theorem for a large class of exponents. Richard Dedekind turned Kummer's ideal numbers into *ideals*, one of the central concepts of modern algebra.

A smaller example to remember

In the number system $\mathbb{Z}[\sqrt{-5}] = \{a + b\sqrt{-5} : a, b \in \mathbb{Z}\}$,

$$6 = 2 \times 3 = (1+\sqrt{-5})(1-\sqrt{-5}),$$

and all four factors are "prime" in that system (exercise 3). Compare this with the Gaussian integers of Section 1.6, where factorisation *is* unique. Which systems of this kind have unique factorisation is a deep question. Gauss conjectured that, among systems of the form $\mathbb{Z}[\sqrt{-d}]$ and their close relatives, only nine have it. This was finally proved by Heegner, Baker and Stark between 1952 and 1967.

Try this next

1 **Another example.** Show that 693 has two different factorisations into S-primes. Can you find a general recipe for producing such examples?

2 **Even numbers.** Let $E = \{2, 4, 6, 8, \ldots\}$. Which numbers are E-prime? Show that 60 has more than one factorisation into E-primes.

3 $\mathbb{Z}[\sqrt{-5}]$**.** Define the *norm* $N(a + b\sqrt{-5}) = a^2 + 5b^2$. Show that $N(zw) = N(z)N(w)$. By showing that no element has norm 2 or 3, deduce that 2, 3 and $1 \pm \sqrt{-5}$ cannot be factorised further in $\mathbb{Z}[\sqrt{-5}]$.

4 **Euclid's lemma in action.** HARDER Assuming Euclid's lemma (if a prime divides ab, it divides a or b), prove that every integer greater than 1 has only one factorisation into primes, apart from the order of the factors.

Further reading

- **Ian Stewart & David Tall,** ***Algebraic Number Theory and Fermat's Last Theorem*** (CRC Press). The opening chapters tell the story of Lamé, Kummer and ideals at a level a keen first-year student can follow.
- **John Stillwell,** ***Elements of Number Theory***. Unique factorisation in the integers and in the Gaussian integers, compared side by side.
- **Harold Edwards,** ***Fermat's Last Theorem: A Genetic Introduction*** (Springer). For the ambitious: Kummer's work explained through its history.

The full guided solution is in Section 1.7:
https://beyondstep.keithmcnulty.org/ch01/factorisation-fails.html